

LAPORAN PRAKTIKUM PEMROGRAMAN WEB

Modul 8: CodeIgniter 3 + HMVC + REST API + CRUD AJAX Tanpa Refresh



Oleh:

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A. TUJUAN PRAKTIKUM

Setelah menyelesaikan modul ini, peserta diharapkan mampu:

1. Memahami konsep HMVC pada CodeIgniter 3.
2. Mengimplementasikan struktur modular (feature-based).
3. Membuat REST API berbasis JSON.
4. Mengintegrasikan Model dengan REST Controller.
5. Menggunakan AJAX untuk komunikasi asynchronous.
6. Membuat fitur CRUD tanpa refresh halaman.
7. Menguji API menggunakan Postman.
8. Mengintegrasikan frontend dengan REST API

B. ALAT DAN BAHAN

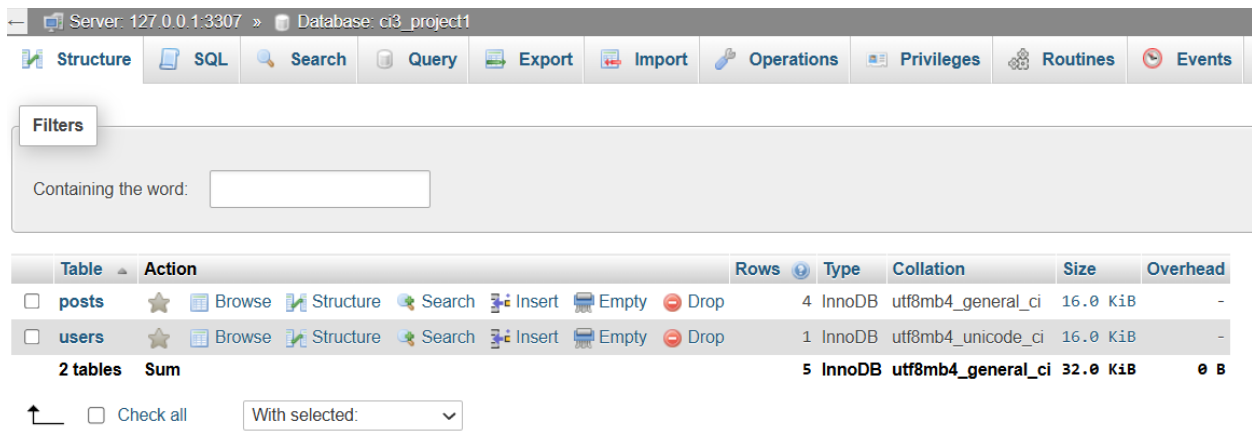
Untuk melaksanakan praktikum ini, diperlukan:

1. XAMPP (Apache & MySQL aktif)
2. PHP 7.x
3. CodeIgniter 3
4. Library HMVC (Modular Extensions – MX)
5. Library REST Server (chriskacerguis)
6. Postman
7. VS Code / Code Editor
8. Browser (Chrome/Firefox)

C. LANGKAH LANGKAH PRAKTIKUM

1) Studi Kasus 1

- a) Create database -> buat table users -> masukkan data dan tipenya, hingga tampilannya seperti ini



The screenshot shows the phpMyAdmin interface for a database named 'ci3_project1'. The 'Structure' tab is selected, displaying the table 'users'. The table has 1 row and 1 column. The column is named 'users' and has a data type of 'InnoDB' with a collation of 'utf8mb4_unicode_ci'. The table size is 16.0 KiB. The interface also shows a 'Filters' section with a search box and a 'Check all' checkbox.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ posts		4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ users		1	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
2 tables	Sum	5	InnoDB	utf8mb4_general_ci	32.0 KiB	0 B

☐ Check all With selected:

Gambar C.1.1 Tabel laporan pada database navibiz

b) Lalu buat code nya di VS Code

```
52 $route['default_controller'] = 'posts';
53 $route['404_override'] = '';
54 $route['translate_uri_dashes'] = FALSE;
55
56 // ===== Auth Routes =====
57 $route['api/auth/register'] = 'api/Auth/register';
58 $route['api/auth/login'] = 'api/Auth/login';
59 $route['api/auth/logout'] = 'api/Auth/logout';
60 $route['api/auth/me'] = 'api/Auth/me';
61
62 // ===== Post Routes =====
63 $route['api/post'] = 'api/Post/handle';
64 $route['api/post/(:num)'] = 'api/Post/handle/$1';
```

Gambar C.1.2 Code pada file routes.php

```
application > libraries > Jwt.php
1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3 class Jwt {
4
5     private $secret_key = 'your_secret_key';
6     private $algorithm = 'HS256';
7     private $expiration = 86400;
8
9     public function __construct()
10     {
11         $CI =& get_instance();
12         $this->secret_key = config_item('jwt_secret_key') ? $this->secret_key;
13         $this->expiration = config_item('jwt_expiration') ? $this->expiration;
14     }
15
16     public function create($payload = array())
17     {
18         $header = array(
19             'typ' => 'JWT',
20             'alg' => $this->algorithm
21         );
22
23         $payload['iat'] = time();
24         $payload['exp'] = time() + $this->expiration;
25
26         $header_encoded = $this->base64_url_encode(json_encode($header));
27         $payload_encoded = $this->base64_url_encode(json_encode($payload));
28
29         $signature = hash_hmac(
30             'sha256',
31             $header_encoded . '.' . $payload_encoded,
32             $this->secret_key,
33             true
34         );
35
36         $signature_encoded = $this->base64_url_encode($signature);
37
38         return $header_encoded . '.' . $payload_encoded . '.' . $signature_encoded;
39     }
40
41     public function verify($token = '')
42     {
43         if (empty($token)) {
44             return false;
45         }
46
47         $parts = explode('.', $token);
48         if (count($parts) != 3) {
49             return false;
50         }
51
52         list($header_encoded, $payload_encoded, $signature_encoded) = $parts;
53
54         $signature = hash_hmac(
55             'sha256',
56             $header_encoded . '.' . $payload_encoded,
```

Gambar C.1.3 Code pada file Jwt.php

```

application > modules > Api > controllers > Auth.php
1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3
4 class Auth extends MY_Controller {
5
6     public function __construct()
7     {
8         parent::__construct();
9         $this->load->model('User_model');
10        $this->load->library('jwt');
11        $this->load->library('form_validation');
12    }
13
14    public function register()
15    {
16        header('Content-Type: application/json; charset=utf-8');
17        ob_clean();
18
19        try {
20            $input = json_decode(file_get_contents('php://input'), true);
21
22            if (empty($input['name']) || empty($input['email']) || empty($input['password'])) {
23                http_response_code(400);
24                echo json_encode([
25                    'status' => false,
26                    'message' => 'Name, email, and password are required'
27                ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
28                exit;
29            }

```

```

application > modules > Api > controllers > Auth.php
4 class Auth extends MY_Controller {
15 {
19     try {
31         if (!filter_var($input['email'], FILTER_VALIDATE_EMAIL)) {
32             http_response_code(400);
33             echo json_encode([
34                 'status' => false,
35                 'message' => 'Invalid email format'
36             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
37             exit;
38         }
39
40         if ($this->User_model->email_exists($input['email'])) {
41             http_response_code(400);
42             echo json_encode([
43                 'status' => false,
44                 'message' => 'Email already registered'
45             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
46             exit;
47         }
48
49         if (strlen($input['password']) < 6) {
50             http_response_code(400);
51             echo json_encode([
52                 'status' => false,
53                 'message' => 'Password must be at least 6 characters'
54             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
55             exit;
56         }

```

```

application > modules > Api > controllers > Auth.php
4 class Auth extends MY_Controller {
15 {
19     try {
58         $user_data = [
59             'email' => strtolower($input['email']),
60             'password' => password_hash($input['password'], PASSWORD_BCRYPT)
61         ];
62
63         $user_id = $this->User_model->create($user_data);
64
65         if (!$user_id) {
66             http_response_code(500);
67             echo json_encode([
68                 'status' => false,
69                 'message' => 'Failed to create user'
70             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
71             exit;
72         }
73
74         $user = $this->User_model->get_by_id($user_id);
75
76         $token = $this->jwt->create([
77             'id' => $user->id,
78             'name' => $user->name,
79             'email' => $user->email
80         ]);
81
82         http_response_code(201);
83         echo json_encode([

```

```

application > modules > Api > controllers > Auth.php
4 class Auth extends MY_Controller {
15 {
19     try {
84         echo json_encode([
85             'status' => true,
86             'message' => 'User registered successfully',
87             'data' => [
88                 'id' => $user->id,
89                 'name' => $user->name,
90                 'email' => $user->email,
91                 'created_at' => $user->created_at
92             ],
93             'access_token' => $token,
94             'token_type' => 'Bearer'
95         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
96         exit;
97     } catch (Exception $e) {
98         http_response_code(500);
99         echo json_encode([
100             'status' => false,
101             'message' => 'Server error: ' . $e->getMessage()
102         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
103         exit;
104     }
105 }
106
107 public function login()
108 {
109

```

```

application > modules > Api > controllers > Auth.php
4 class Auth extends MY_Controller {
109 {
110     header('Content-Type: application/json; charset=utf-8');
111     ob_clean();
112
113     try {
114         $input = json_decode(file_get_contents('php://input'), true);
115
116         if (empty($input['email']) || empty($input['password'])) {
117             http_response_code(400);
118             echo json_encode([
119                 'status' => false,
120                 'message' => 'Email and password are required'
121             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
122             exit;
123         }
124
125         $user = $this->User_model->get_by_email(strtolower($input['email']));
126
127         if (!$user) {
128             http_response_code(401);
129             echo json_encode([
130                 'status' => false,
131                 'message' => 'Invalid credentials'
132             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
133             exit;
134         }
135
136         if (!password_verify($input['password'], $user->password)) {

```

```

application > modules > Api > controllers > Auth.php
4 class Auth extends MY_Controller {
109 {
113     try {
116         if (!password_verify($input['password'], $user->password)) {
117             http_response_code(401);
118             echo json_encode([
119                 'status' => false,
120                 'message' => 'Invalid credentials'
121             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
122             exit;
123         }
124
125         $token = $this->jwt->create([
126             'id' => $user->id,
127             'name' => $user->name,
128             'email' => $user->email
129         ]);
130
131         http_response_code(200);
132         echo json_encode([
133             'status' => true,
134             'message' => 'Login successful',
135             'data' => [
136                 'id' => $user->id,
137                 'name' => $user->name,
138                 'email' => $user->email
139             ],
140             'access_token' => $token,
141             'token_type' => 'Bearer'
142         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
143         exit;
144     }

```

```

application > modules > Api > controllers > Auth.php
4      class Auth extends MY_Controller {
176
177          {
180
181              try {
192                  $decoded = $this->jwt->verify($token);
193
194                  if ( ! $decoded ) {
195                      http_response_code(401);
196                      echo json_encode([
197                          'status' => false,
198                          'message' => 'Invalid token'
199                      ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
200                      exit;
201                  }
202
203                  $user = $this->User_model->get_by_id($decoded->id);
204
205                  if ( !$user ) {
206                      http_response_code(404);
207                      echo json_encode([
208                          'status' => false,
209                          'message' => 'User not found'
210                      ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
211                      exit;
212                  }
213
214                  http_response_code(200);
215                  echo json_encode([
216                      'status' => true,
217                      'message' => 'Logout successful'

```

```

4 application > modules > Api > controllers > Auth.php
class Auth extends MY_Controller {
232     {
233     try {
234         if ($token) {
235             echo json_encode([
236                 ], 350N_UNESCAPED_SLASHES | 350N_UNESCAPED_UNICODIC);
237             exit;
238         }
239     }
240     $decoded = $this->jwt->verify($token);
241     if ($decoded) {
242         http_response_code(401);
243         echo json_encode([
244             'status' => false,
245             'message' => 'Invalid or expired token'
246         ], 350N_UNESCAPED_SLASHES | 350N_UNESCAPED_UNICODIC);
247         exit;
248     }
249     $user = $this->User_model->get_by_id($decoded->id);
250     if ($user) {
251         http_response_code(404);
252         echo json_encode([
253             'status' => false,
254             'message' => 'User not found'
255         ], 350N_UNESCAPED_SLASHES | 350N_UNESCAPED_UNICODIC);
256         exit;
257     }
258 }
259 }
260 }
261 }
262 }
263 }
264 }
265 }
266 }
267 }

```

```

application > modules > Api > controllers > @Auth.php
4 class Auth extends MY_Controller {
232     {
233         try {
269
270             http_response_code(200);
271             echo json_encode([
272                 'status' => true,
273                 'message' => 'User retrieved successfully',
274                 'data' => [
275                     'id' => $user->id,
276                     'name' => $user->name,
277                     'email' => $user->email,
278                     'created_at' => $user->created_at,
279                     'updated_at' => $user->updated_at
280                 ]
281             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
282             exit;
283
284         } catch (Exception $e) {
285             http_response_code(500);
286             echo json_encode([
287                 'status' => false,
288                 'message' => 'Server error: ' . $e->getMessage()
289             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
290             exit;
291         }
292     }
293 }
294

```

Gambar C.1.4 application/modules/Api/controllers/Auth.php

```

application > modules > Api > controllers > Post.php
1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3
4 class Post extends MX_Controller {
5
6     public function __construct()
7     {
8         parent::__construct();
9         $this->load->model('Post_model');
10        $this->load->library('jwt');
11        $this->load->library('upload');
12    }
13
14    private function parse_multipart_form()
15    {
16        $content_type = $_SERVER['CONTENT_TYPE'] ?? '';
17        $input = [];
18
19        if (strpos($content_type, 'multipart/form-data') !== false) {
20            $boundary = '';
21            if (preg_match('/boundary=([^\s]+)/', $content_type, $matches)) {
22                $boundary = trim($matches[1], '"');
23            }
24
25            if ($boundary) {
26                return [];
27            }
28
29            $raw = file_get_contents('php://input');

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
5
6     {
7         if (strpos($content_type, 'multipart/form-data') !== false) {
8             $parts = preg_split('/' . preg_quote('--$boundary') . '/', $raw);
9
10            foreach ($parts as $part) {
11                if (empty(trim($part)) || $part === "--\n\n" || $part === "--") {
12                    continue;
13                }
14
15                $split = preg_split("/\n\n\r\n/", trim($part), 2);
16                if (count($split) !== 2) {
17                    continue;
18                }
19
20                $headers = $split[0];
21                $content = rtrim($split[1], "\n\n");
22
23                if (preg_match('/name=("[^\s"]+)/', $headers, $matches)) {
24                    $name = $matches[1];
25
26                    if (preg_match('/filename=("[^\s"]+)/', $headers, $filename_matches)) {
27                        $file_content_type = 'application/octet-stream';
28                        if (preg_match('/Content-Type: ([^\s]+)/', $headers, $type_matches)) {
29                            $file_content_type = trim($type_matches[1]);
30                        }
31
32                        $input[$name] = [
33                            'name' => $filename_matches[1],

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
5
6     {
7         if (strpos($content_type, 'multipart/form-data') !== false) {
8             foreach ($parts as $part) {
9                 if (preg_match('/name=("[^\s"]+)/', $headers, $matches)) {
10                     } else {
11                         $input[$name] = $content;
12                     }
13                 }
14             }
15
16            return $input;
17
18            if (strpos($content_type, 'application/x-www-form-urlencoded') !== false) {
19                parse_str(file_get_contents('php://input'), $input);
20                return $input ?? [];
21            }
22
23            return array_merge($_POST ?? [], $_FILES ?? []);
24        }
25
26        private function get_image_url($image_filename)
27        {
28            if (empty($image_filename)) {
29                return null;
30            }
31
32            return 'http://localhost/cis_project1/uploads/posts/' . $image_filename;
33        }

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
5
6     {
7         $method = $_SERVER['REQUEST_METHOD'];
8
9         if ($method === 'GET') {
10             if ($id) {
11                 $this->show($id);
12             } else {
13                 $this->index();
14             }
15
16             } elseif ($method === 'POST') {
17                 $this->store();
18             } elseif ($method === 'PUT') {
19                 $this->update($id);
20             } elseif ($method === 'DELETE') {
21                 $this->delete($id);
22             } else {
23                 http_response_code(405);
24                 header('Content-Type: application/json; charset=utf-8');
25                 echo json_encode([
26                     'status' => false,
27                     'message' => 'Method not allowed'
28                 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
29                 exit;
30             }
31         }
32     }

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
5
6     public function index()
7     {
8         header('Content-Type: application/json; charset=utf-8');
9         ob_clean();
10
11         try {
12             $page = isset($_GET['page']) ? max(1, (int)$_GET['page']) : 1;
13             $per_page = isset($_GET['per_page']) ? max(1, min(100, (int)$_GET['per_page'])) : 10;
14             $offset = ($page - 1) * $per_page;
15
16             $posts = $this->Post_model->get_all($per_page, $offset);
17             $total = $this->Post_model->count_all();
18
19             foreach ($posts as $post) {
20                 if (empty($post->image)) {
21                     $post->image_url = $this->get_image_url($post->image);
22                 } else {
23                     $post->image_url = null;
24                 }
25             }
26
27             http_response_code(200);
28             echo json_encode([
29                 'status' => true,
30                 'message' => 'Posts retrieved successfully',
31                 'data' => $posts,
32                 'pagination' => [
33                     'total' => (int)$total,

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
5
6     {
7         try {
8             echo json_encode([
9                 'pagination' => [
10                     'total' => (int)$total,
11                     'per_page' => $per_page,
12                     'current_page' => $page,
13                     'last_page' => ceil($total / $per_page)
14                 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
15             exit;
16         } catch (Exception $e) {
17             http_response_code(500);
18             echo json_encode([
19                 'status' => false,
20                 'message' => 'Server error: ' . $e->getMessage()
21             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
22             exit;
23         }
24     }
25
26     public function show($id)
27     {
28         header('Content-Type: application/json; charset=utf-8');
29         ob_clean();
30
31         try {

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
157 {
161     try {
162         if (!is_numeric($id)) {
163             http_response_code(400);
164             echo json_encode([
165                 'status' => false,
166                 'message' => 'Invalid post ID'
167             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
168             exit;
169         }
170
171         $post = $this->Post_model->get_by_id($id);
172
173         if (!$post) {
174             http_response_code(404);
175             echo json_encode([
176                 'status' => false,
177                 'message' => 'Post not found'
178             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
179             exit;
180         }
181
182         if (empty($post->image)) {
183             $post->image_url = $this->get_image_url($post->image);
184         } else {
185             $post->image_url = null;
186         }
187
188         http_response_code(200);

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
157 {
161     try {
162         http_response_code(200);
163         echo json_encode([
164             'status' => true,
165             'message' => 'Post retrieved successfully',
166             'data' => $post
167         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
168         exit;
169     } catch (Exception $e) {
170         http_response_code(500);
171         echo json_encode([
172             'status' => false,
173             'message' => 'Server error: ' . $e->getMessage()
174         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
175         exit;
176     }
177 }
178
179 public function store()
180 {
181     header('Content-Type: application/json; charset=utf-8');
182     ob_clean();
183
184     try {
185         $token = $this->get_token_from_request();

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
207 {
211     try {
212         if (!$token) {
213             http_response_code(401);
214             echo json_encode([
215                 'status' => false,
216                 'message' => 'Unauthorized: No token provided'
217             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
218             exit;
219         }
220
221         $decoded = $this->jwt->verify($token);
222
223         if (!$decoded) {
224             http_response_code(401);
225             echo json_encode([
226                 'status' => false,
227                 'message' => 'Unauthorized: Invalid or expired token'
228             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
229             exit;
230         }
231
232         $input = json_decode(file_get_contents('php://input'), true);
233
234         if (!$input || empty($POST)) {
235             $input = $POST;
236         }
237
238         if ($input && !empty($POST)) {

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
207 {
211     try {
212         if (empty($input['title']) || empty($input['article'])) {
213             http_response_code(400);
214             echo json_encode([
215                 'status' => false,
216                 'message' => 'Title and article are required'
217             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
218             exit;
219         }
220
221         $post_data = [
222             'title' => htmlspecialchars($input['title']),
223             'author' => htmlspecialchars($input['author'] ?? 'Anonymous'),
224             'article' => htmlspecialchars($input['article'])
225         ];
226
227         $image_name = '';
228         if (empty($_FILES['image']['name'])) {
229             $image_name = $this->upload_image();
230         }
231         if ($image_name) {
232             http_response_code(400);
233             echo json_encode([
234                 'status' => false,
235                 'message' => 'Image upload failed'
236             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
237             exit;
238         }
239     }
240 }

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
207 {
211     try {
212         if (!$post_id) {
213             http_response_code(500);
214             echo json_encode([
215                 'status' => false,
216                 'message' => 'Failed to create post'
217             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
218             exit;
219         }
220
221         $post = $this->Post_model->get_by_id($post_id);
222
223         if (empty($post->image)) {
224             $post->image_url = $this->get_image_url($post->image);
225         } else {
226             $post->image_url = null;
227         }
228
229         http_response_code(201);
230         echo json_encode([
231             'status' => true,
232             'message' => 'Post created successfully',
233             'data' => $post
234         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
235         exit;
236     }
237 }

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
207 {
211     public function update($id)
212     {
213         header('Content-Type: application/json; charset=utf-8');
214         ob_clean();
215
216         try {
217             $token = $this->jwt->get_token_from_request();
218
219             if (!$token) {
220                 http_response_code(401);
221                 echo json_encode([
222                     'status' => false,
223                     'message' => 'Unauthorized: No token provided'
224                 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
225                 exit;
226             }
227
228             $decoded = $this->jwt->verify($token);
229
230             if (!$decoded) {
231                 http_response_code(401);
232                 echo json_encode([
233                     'status' => false,
234                     'message' => 'Unauthorized: Invalid or expired token'
235                 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
236                 exit;
237             }
238         }
239     }
240 }

```



```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
387 {
388     try {
389         if (!is_numeric($id)) {
390             http_response_code(400);
391             echo json_encode([
392                 'status' => false,
393                 'message' => 'Invalid post ID'
394             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
395             exit;
396         }
397         if (!$this->Post_model->exists($id)) {
398             http_response_code(404);
399             echo json_encode([
400                 'status' => false,
401                 'message' => 'Post not found'
402             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
403             exit;
404         }
405         $input = json_decode(file_get_contents('php://input'), true);
406         if ($input) {
407             $input = $this->parse_multipart_form();
408             if (!empty($input['image']) && is_array($input['image']) && isset($input['image'])['cc']) {
409                 $FILES['image'] = [
410                     'name' => $input['image']['name'] ?? 'image_' . time() . '.png',
411                     'type' => $input['image']['type'] ?? 'image/png',

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
387 {
388     try {
389         if (empty($input['title']) || empty($input['article'])) {
390             http_response_code(400);
391             echo json_encode([
392                 'status' => false,
393                 'message' => 'Title and article are required'
394             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
395             exit;
396         }
397         $post_data = [
398             'title' => htmlspecialchars($input['title']),
399             'author' => htmlspecialchars($input['author'] ?? 'Anonymous'),
400             'article' => htmlspecialchars($input['article'])
401         ];
402         if (empty($_FILES['image']['name']) || (isset($input['image']) && is_array($input['image'])
403             $old_post = $this->Post_model->get_by_id($id);
404             // Delete old image
405             if (empty($old_post->image)) {
406                 $old_file = FCPATH . 'uploads/posts/' . $old_post->image;
407                 if (file_exists($old_file)) {
408                     unlink($old_file);
409                 }
410             }

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
387 {
388     try {
389         if (empty($_FILES['image']['name']) || (isset($input['image']) && is_array($input['image'])
390             $image_name = $this->upload_image();
391             if ($image_name) {
392                 $post_data['image'] = $image_name;
393             }
394             $result = $this->Post_model->update($id, $post_data);
395             if (!$result) {
396                 http_response_code(500);
397                 echo json_encode([
398                     'status' => false,
399                     'message' => 'Failed to update post'
400                 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
401                 exit;
402             }
403             $post = $this->Post_model->get_by_id($id);
404             if (empty($post->image)) {
405                 $post->image_url = $this->get_image_url($post->image);
406             } else {
407                 $post->image_url = null;
408             }
409         }
410     }
411 }
412
413
414
415
416
417

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
387 {
388     try {
417         http_response_code(200);
418         echo json_encode([
419             'status' => true,
420             'message' => 'Post updated successfully',
421             'data' => $post
422         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
423         exit;
424     } catch (Exception $e) {
425         http_response_code(500);
426         echo json_encode([
427             'status' => false,
428             'message' => 'Server error: ' . $e->getMessage()
429         ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
430         exit;
431     }
432 }
433
434 public function delete($id)
435 {
436     header('Content-Type: application/json; charset=utf-8');
437     ob_clean();
438     try {
439         $token = $this->jwt->get_token_from_request();

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
437 {
438     try {
439         $token = $this->jwt->get_token_from_request();
440         if (!$token) {
441             http_response_code(401);
442             echo json_encode([
443                 'status' => false,
444                 'message' => 'Unauthorized: No token provided'
445             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
446             exit;
447         }
448         $decoded = $this->jwt->verify($token);
449         if (!$decoded) {
450             http_response_code(401);
451             echo json_encode([
452                 'status' => false,
453                 'message' => 'Unauthorized: Invalid or expired token'
454             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
455             exit;
456         }
457         if (!is_numeric($id)) {
458             http_response_code(400);
459             echo json_encode([
460                 'status' => false,

```

```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
437 {
438     try {
464         if (!is_numeric($id)) {
465             http_response_code(400);
466             echo json_encode([
467                 'status' => false,
468                 'message' => 'Invalid post ID'
469             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
470             exit;
471         }
472         $post = $this->Post_model->get_by_id($id);
473         if (!$post) {
474             http_response_code(404);
475             echo json_encode([
476                 'status' => false,
477                 'message' => 'Post not found'
478             ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
479             exit;
480         }
481         if (empty($post->image)) {
482             $file = FCPATH . 'uploads/posts/' . $post->image;
483             if (file_exists($file)) {
484                 unlink($file);
485             }
486         }
487         $result = $this->Post_model->delete($id);

```



```

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
437 {
441 try {
493 if (!$result) {
494 http_response_code(500);
495 echo json_encode([
496 'status' => false,
497 'message' => 'Failed to delete post'
498 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
499 exit;
500 }
501 http_response_code(200);
502 echo json_encode([
503 'status' => true,
504 'message' => 'Post deleted successfully'
505 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
506 exit;
507 } catch (Exception $e) {
508 http_response_code(500);
509 echo json_encode([
510 'status' => false,
511 'message' => 'Server error: ' . $e->getMessage()
512 ], JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
513 exit;
514 }
515 }
516 }
517 }

application > modules > Api > controllers > Post.php
4 class Post extends MX_Controller {
519 private function _save_temp_file($content)
520 {
521 $temp_path = sys_get_temp_dir() . '/c13_upload_' . uniqid() . '.tmp';
522 file_put_contents($temp_path, $content);
523 return $temp_path;
524 }
525 private function _upload_image()
526 {
527 $config['upload_path'] = FCPATH . 'uploads/posts/';
528 $config['allowed_types'] = 'gif|jpg|png|jpeg';
529 $config['max_size'] = 2048; // 2MB
530 $config['file_name'] = 'post_' . time() . '_' . uniqid();
531 // Create directory if not exists
532 if (!is_dir($config['upload_path'])) {
533 mkdir($config['upload_path'], 0755, true);
534 }
535 $this->upload->initialize($config);
536 if (!$this->upload->do_upload('image')) {
537 return false;
538 }
539 $upload_data = $this->upload->data();
540 return $upload_data['file_name'];
541 }
542 }
543 }
544 }
545 }
546 }

```

Gambar C.1.5 application/modules /Api/controllers/Post.php

```

application > modules > Api > models > Post_model.php
1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3
4 class Post_model extends CI_Model {
5
6 private $table = 'posts';
7
8 public function __construct()
9 {
10 parent::__construct();
11 $this->load->database();
12 }
13
14 public function get_all()
15 {
16 $query = $this->db->get($this->table);
17 $records = $query->result();
18
19 foreach ($records as $record) {
20 if (empty($record->image)) {
21 $record->image_url = base_url('uploads/posts/' . $record->image);
22 } else {
23 $record->image_url = null;
24 }
25 }
26
27 return $records;
28 }
29 }

application > modules > Api > models > Post_model.php
4 class Post_model extends CI_Model {
30 public function count_all()
31 {
32 return $this->db->count_all($this->table);
33 }
34
35 public function get_by_id($id)
36 {
37 $query = $this->db
38 ->select('id, title, author, article, image, created_at, updated_at')
39 ->get_where($this->table, array('id' => $id));
40 return $query->row();
41 }
42
43 public function create($data)
44 {
45 $data['created_at'] = date('Y-m-d H:i:s');
46 $data['updated_at'] = date('Y-m-d H:i:s');
47
48 $this->db->insert($this->table, $data);
49 return $this->db->insert_id();
50 }
51
52 public function update($id, $data)
53 {
54 $data['updated_at'] = date('Y-m-d H:i:s');
55 $this->db->where('id', $id);
56 return $this->db->update($this->table, $data);
57 }
58 }

```

```

application > modules > Api > models > Post_model.php
4 class Post_model extends CI_Model {
59 public function delete($id)
60 {
61 $this->db->where('id', $id);
62 return $this->db->delete($this->table);
63 }
64
65 public function exists($id)
66 {
67 $query = $this->db->get_where($this->table, array('id' => $id));
68 return $query->num_rows() > 0;
69 }
70
71 public function search($keyword, $limit = 10, $offset = 0)
72 {
73 $this->db->like('title', $keyword);
74 $this->db->or_like('description', $keyword);
75 $this->db->limit($limit, $offset);
76 $this->db->order_by('created_at', 'DESC');
77 $query = $this->db->get($this->table);
78 return $query->result();
79 }
80
81 public function get_by_user($user_id, $limit = 10, $offset = 0)
82 {
83 $query = $this->db
84 ->select('id, title, author, article, image, created_at, updated_at')
85 ->where('user_id', $user_id)
86 ->limit($limit, $offset)
87 ->order_by('created_at', 'DESC');
88 }
89 }

```

Gambar C.1.6 application/modules/modules/Api/models/Post_Moduls.php

```

application > modules > Api > models > User_model.php
1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3
4 class User_model extends CI_Model {
5
6     private $table = 'users';
7
8     public function __construct()
9     {
10         parent::__construct();
11         $this->load->database();
12     }
13
14     public function get_all()
15     {
16         $query = $this->db->get($this->table);
17         return $query->result();
18     }
19
20     public function get_by_id($id)
21     {
22         $query = $this->db->get_where($this->table, array('id' => $id));
23         return $query->row();
24     }
25
26     public function get_by_email($email)
27     {
28         $query = $this->db->get_where($this->table, array('email' => $email));
29         return $query->row();
30     }
31 }

```

```

application > modules > Api > models > User_model.php
4 class User_model extends CI_Model {
5
6     public function create($data)
7     {
8         $data['created_at'] = date('Y-m-d H:i:s');
9         $data['updated_at'] = date('Y-m-d H:i:s');
10
11         $this->db->insert($this->table, $data);
12         return $this->db->insert_id();
13     }
14
15     public function update($id, $data)
16     {
17         $data['updated_at'] = date('Y-m-d H:i:s');
18         $this->db->where('id', $id);
19         return $this->db->update($this->table, $data);
20     }
21
22     public function delete($id)
23     {
24         $this->db->where('id', $id);
25         return $this->db->delete($this->table);
26     }
27
28     public function email_exists($email)
29     {
30         $query = $this->db->get_where($this->table, array('email' => $email));
31         return $query->num_rows() > 0;
32     }
33 }

```

```

60     public function exists($id)
61     {
62         $query = $this->db->get_where($this->table, array('id' => $id));
63         return $query->num_rows() > 0;
64     }
65 }

```

Gambar C.1.7 application/modules/modules/Api/models/User_Moduls.php

```

application > modules > Crudjs > controllers > Crudjs.php
1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3
4 class Crudjs extends CI_Controller {
5
6     public function __construct()
7     {
8         parent::__construct();
9         $this->load->model('Crudjs_model');
10         $this->load->helper(array('url', 'form'));
11         $this->load->library(array('form_validation', 'upload', 'session'));
12     }
13
14     public function debug()
15     {
16         $debug_info = array();
17
18         $debug_info['database'] = array(
19             'connected' => $this->db->conn_id ? 'Yes' : 'No',
20             'database' => $this->db->database
21         );
22
23         $debug_info['table_exists'] = $this->db->table_exists('posts') ? 'Yes' : 'No';
24
25         $query = $this->db->get('posts', 1);
26         $debug_info['table_accessible'] = $query ? 'Yes' : 'No';
27         $debug_info['record_count'] = $this->db->count_all('posts');
28
29         $upload_path = './uploads/posts/';
30     }
31 }

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
5
6     public function debug()
7     {
8         $debug_info['upload_folder'] = array(
9             'exists' => is_dir($upload_path) ? 'Yes' : 'No',
10             'writable' => is_writable($upload_path) ? 'Yes' : 'No'
11         );
12
13         $debug_info['sample_records'] = array();
14         try {
15             $records = $this->Crudjs_model->get_all();
16             if (empty($records)) {
17                 $first = $records[0];
18                 $debug_info['sample_records'] = array(
19                     'count' => count($records),
20                     'first_record' => $first
21                 );
22             }
23         } catch (Exception $e) {
24             $debug_info['sample_records']['error'] = $e->getMessage();
25         }
26
27         echo '<pre>'. json_encode($debug_info, JSON_PRETTY_PRINT | JSON_UNESCAPED_SLASHES) . '</pre>';
28     }
29
30     public function index()
31     {
32         $data['title'] = 'CRUD with AJAX';
33         $this->load->view('crudjs/index', $data);
34     }
35 }

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
5
6     public function index()
7     {
8         $data['title'] = 'CRUD with AJAX';
9         $this->load->view('crudjs/index', $data);
10     }
11
12     public function get_all()
13     {
14         ob_clean();
15         header('Content-Type: application/json; charset=utf-8');
16
17         try {
18             $records = $this->Crudjs_model->get_all();
19
20             $response = array(
21                 'status' => 'success',
22                 'data' => $records
23             );
24
25             echo json_encode($response, JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
26         } catch (Exception $e) {
27             echo json_encode(array(
28                 'status' => 'error',
29                 'message' => 'Failed to load records: ' . $e->getMessage()
30             ));
31         }
32         exit;
33     }
34 }

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
5
6     public function get_record($id)
7     {
8         ob_clean();
9         header('Content-Type: application/json; charset=utf-8');
10
11         try {
12             if ($this->Crudjs_model->exists($id)) {
13                 echo json_encode(array(
14                     'status' => 'error',
15                     'message' => 'Record not found'
16                 ));
17                 exit;
18             }
19
20             $record = $this->Crudjs_model->get_by_id($id);
21
22             echo json_encode(array(
23                 'status' => 'success',
24                 'data' => $record
25             ), JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
26         } catch (Exception $e) {
27             echo json_encode(array(
28                 'status' => 'error',
29                 'message' => 'Failed to load record: ' . $e->getMessage()
30             ));
31         }
32         exit;
33     }
34 }

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
110 public function store()
111 {
112     ob_clean();
113     header('Content-Type: application/json; charset=utf-8');
114
115     try {
116         $this->form_validation->set_rules('title', 'Title', 'required|max_length[255]');
117         $this->form_validation->set_rules('author', 'Author', 'required|max_length[255]');
118         $this->form_validation->set_rules('article', 'Article', 'required');
119
120         if ($this->form_validation->run() === FALSE) {
121             echo json_encode(array(
122                 'status' => 'error',
123                 'message' => 'Validation error: ' . strip_tags(validation_errors())
124             ));
125             exit;
126         }
127
128         $data = array(
129             'title' => $this->input->post('title'),
130             'author' => $this->input->post('author'),
131             'article' => $this->input->post('article'),
132             'created_at' => date('Y-m-d H:i:s'),
133             'updated_at' => date('Y-m-d H:i:s')
134         );
135
136         if (empty($_FILES['image']['name'])) {
137             $upload_result = $this->upload_image();

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
111 {
112     try {
113         if (empty($_FILES['image']['name'])) {
114             if ($upload_result['status']) {
115                 $data['image'] = 'posts/' . $upload_result['file_name'];
116             } else {
117                 echo json_encode(array(
118                     'status' => 'error',
119                     'message' => $upload_result['error']
120                 ));
121                 exit;
122             }
123         }
124
125         $record_id = $this->Crudjs_model->insert($data);
126
127         if ($record_id) {
128             $new_record = $this->Crudjs_model->get_by_id($record_id);
129             echo json_encode(array(
130                 'status' => 'success',
131                 'message' => 'Record created successfully!',
132                 'data' => $new_record
133             ), JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
134         } else {
135             echo json_encode(array(
136                 'status' => 'error',
137                 'message' => 'Failed to create record'
138             ));
139         }
140     }
141 }

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
111 {
112     try {
113         if (empty($_FILES['image']['name'])) {
114             if ($upload_result['status']) {
115                 $data['image'] = 'posts/' . $upload_result['file_name'];
116             } else {
117                 echo json_encode(array(
118                     'status' => 'error',
119                     'message' => 'Failed to create record'
120                 ));
121             }
122         } catch (Exception $e) {
123             echo json_encode(array(
124                 'status' => 'error',
125                 'message' => 'Exception error: ' . $e->getMessage()
126             ));
127             exit;
128         }
129     }
130 }
131
132 public function update($id)
133 {
134     ob_clean();
135     header('Content-Type: application/json; charset=utf-8');
136
137     try {
138         if (!$this->Crudjs_model->exists($id)) {
139             echo json_encode(array(
140                 'status' => 'error',
141                 'message' => 'Record not found'
142             ));
143             exit;
144         }
145     }
146 }

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
175 {
176     try {
177         $this->form_validation->set_rules('title', 'Title', 'max_length[255]');
178         $this->form_validation->set_rules('author', 'Author', 'max_length[255]');
179
180         if ($this->form_validation->run() === FALSE) {
181             echo json_encode(array(
182                 'status' => 'error',
183                 'message' => 'Validation error: ' . strip_tags(validation_errors())
184             ));
185             exit;
186         }
187
188         $data = array(
189             'updated_at' => date('Y-m-d H:i:s')
190         );
191
192         if ($this->input->post('title')) {
193             $data['title'] = $this->input->post('title');
194         }
195         if ($this->input->post('author')) {
196             $data['author'] = $this->input->post('author');
197         }
198         if ($this->input->post('article')) {
199             $data['article'] = $this->input->post('article');
200         }
201
202         if (empty($_FILES['image']['name'])) {

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
175 {
176     try {
177         if (empty($_FILES['image']['name'])) {
178             $old_record = $this->Crudjs_model->get_by_id($id);
179             if ($old_record->image) {
180                 $old_image_path = './uploads/' . $old_record->image;
181                 if (file_exists($old_image_path)) {
182                     unlink($old_image_path);
183                 }
184             }
185
186             $upload_result = $this->upload_image();
187
188             if ($upload_result['status']) {
189                 $data['image'] = 'posts/' . $upload_result['file_name'];
190             } else {
191                 echo json_encode(array(
192                     'status' => 'error',
193                     'message' => $upload_result['error']
194                 ));
195                 exit;
196             }
197         }
198
199         $result = $this->Crudjs_model->update($id, $data);
200
201         if ($result) {
202             $updated_record = $this->Crudjs_model->get_by_id($id);

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
175 {
176     try {
177         if ($result) {
178             echo json_encode(array(
179                 'status' => 'success',
180                 'message' => 'Record updated successfully!',
181                 'data' => $updated_record
182             ), JSON_UNESCAPED_SLASHES | JSON_UNESCAPED_UNICODE);
183         } else {
184             echo json_encode(array(
185                 'status' => 'error',
186                 'message' => 'Failed to update record'
187             ));
188         }
189     } catch (Exception $e) {
190         echo json_encode(array(
191             'status' => 'error',
192             'message' => 'Exception error: ' . $e->getMessage()
193         ));
194     }
195     exit;
196 }
197
198 public function delete($id)
199 {
200     ob_clean();
201     header('Content-Type: application/json; charset=utf-8');
202

```

```

application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
260 {
261     try {
262         $record = $this->Crudjs_model->get_by_id($id);
263     }
264     if (!$record) {
265         echo json_encode(array(
266             'status' => 'error',
267             'message' => 'Record not found'
268         ));
269         exit;
270     }
271     // Delete image if exists
272     if ($record->image) {
273         $image_path = './uploads/' . $record->image;
274         if (file_exists($image_path)) {
275             unlink($image_path);
276         }
277     }
278     $result = $this->Crudjs_model->delete($id);
279     if ($result) {
280         echo json_encode(array(
281             'status' => 'success',
282             'message' => 'Record deleted successfully!'
283         ));
284     } else {
285
application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
260 {
261     try {
262         $record = $this->Crudjs_model->get_by_id($id);
263     } catch (Exception $e) {
264         echo json_encode(array(
265             'status' => 'error',
266             'message' => 'Exception error: ' . $e->getMessage()
267         ));
268         exit;
269     }
270     // Delete image if exists
271     if ($record->image) {
272         $image_path = './uploads/' . $record->image;
273         if (file_exists($image_path)) {
274             unlink($image_path);
275         }
276     }
277     $result = $this->Crudjs_model->delete($id);
278     if ($result) {
279         echo json_encode(array(
280             'status' => 'success',
281             'message' => 'Record deleted successfully!'
282         ));
283     } else {
284
application > modules > Crudjs > controllers > Crudjs.php
4 class Crudjs extends CI_Controller {
314 {
315     $original_name = $_FILES['image']['name'];
316     $sanitized_name = preg_replace('/[^A-Za-z0-9._-]/', '_', $original_name);
317     $file_name = time() . '_' . $sanitized_name;
318
319     $config['upload_path'] = $upload_path;
320     $config['allowed_types'] = 'jpg|jpeg|png';
321     $config['max_size'] = 2048; // 2MB
322     $config['file_name'] = $file_name;
323     $config['overwrite'] = FALSE;
324
325     $this->upload->initialize($config);
326
327     if ($this->upload->do_upload('image')) {
328         $upload_data = $this->upload->data();
329         return array(
330             'status' => TRUE,
331             'file_name' => $upload_data['file_name']
332         );
333     } else {
334         return array(
335             'status' => FALSE,
336             'error' => $this->upload->display_errors('', '')
337         );
338     }
339 }
340
341

```

Gambar C.1.8 application/modules/Crudjs/controllers/Crudjs.php

```

1 <?php
2 defined('BASEPATH') OR exit('No direct script access allowed');
3
4 class Crudjs_model extends CI_Model {
5
6     private $table = 'posts';
7
8     public function __construct()
9     {
10         parent::__construct();
11         $this->load->database();
12     }
13
14     public function get_all()
15     {
16         $query = $this->db->get($this->table);
17         $records = $query->result();
18
19         foreach ($records as $record) {
20             if ($record->image) {
21                 $record->image_url = 'http://localhost/ci3_project/uploads/posts/' . $record->image;
22             } else {
23                 $record->image_url = null;
24             }
25         }
26
27         return $records;
28     }
29 }

```

```

4 class Crudjs_model extends CI_Model {
5     public function get_by_id($id)
6     {
7         $query = $this->db->get_where($this->table, array('id' => $id));
8         $record = $query->row();
9
10         if ($record && $record->image) {
11             $record->image_url = 'http://localhost/ci3_project/uploads/posts/' . $record->image;
12         } else if ($record) {
13             $record->image_url = null;
14         }
15
16         return $record;
17     }
18
19     public function insert($data)
20     {
21         $this->db->insert($this->table, $data);
22         return $this->db->insert_id();
23     }
24
25     public function update($id, $data)
26     {
27         $this->db->where('id', $id);
28         return $this->db->update($this->table, $data);
29     }
30
31     public function delete($id)
32     {
33         $this->db->where('id', $id);
34         return $this->db->delete($this->table);
35     }
36
37     public function exists($id)
38     {
39         $query = $this->db->get_where($this->table, array('id' => $id));
40         return $query->num_rows() > 0;
41     }
42 }

```

```

44 public function insert($data)
45 {
46     $this->db->insert($this->table, $data);
47     return $this->db->insert_id();
48 }
49
50 public function update($id, $data)
51 {
52     $this->db->where('id', $id);
53     return $this->db->update($this->table, $data);
54 }
55
56 public function delete($id)
57 {
58     $this->db->where('id', $id);
59     return $this->db->delete($this->table);
60 }
61
62 public function exists($id)
63 {
64     $query = $this->db->get_where($this->table, array('id' => $id));
65     return $query->num_rows() > 0;
66 }
67 }

```

Gambar C.1.9 application/modules/Crudjs/models/Crudjs_model.php

```

application > modules > Crudjs > views > footer.php
1 <footer>
2 <div class="container">
3 <p>&copy; <?php echo date('Y'); ?> CRUD with AJAX - Built with CodeIgniter 3</p>
4 </div>
5 </footer>
6
7 <div id="formModal" class="modal">
8 <div class="modal-content">
9 <div class="modal-header">
10 <h2 id="modalTitle">Create New Record</h2>
11 <span class="close" onclick="closeModal()">&times;</span></div>
12 </div>
13 <div class="modal-body">
14 <div class="loading" id="loading">
15 <div class="spinner"></div>
16 <p>Processing...</p>
17 </div>
18 <form id="crudForm" enctype="multipart/form-data">
19 <input type="hidden" id="recordId" name="recordId">
20
21 <div class="form-group">
22 <label for="title">Title <span style="color: red;">*</span></label>
23 <input type="text"
24 <name="title"
25 <id="title"
26 <placeholder="Enter title"
27 <required>
28 </div>
29
application > modules > Crudjs > views > header.php
3 <head>
7 <style>
423 </style>
424 </head>
425 </head>
426 <body>
427 <header>
428 <div class="container">
429 <h1> CRUD with AJAX</h1>
430 <p>CodeIgniter 3 CRUD Application - No Page Refresh</p>
431 </div>
432 </header>
433
434 <div class="container">
435 <div class="nav">
436 <ul>
437 <li><a href="{?php echo base_url(); ?}">Home</a></li>
438 <li><a href="{?php echo base_url('crudjs')}"; ?>CRUD with AJAX</a></li>
439 </ul>
440 </div>
441
442 <div id="alertContainer">
443 <div id="successAlert" class="alert alert-success"></div>
444 <div id="errorAlert" class="alert alert-error"></div>
445 </div>
446
application > modules > Crudjs > views > index.php
1 <?php $this->load->view('crudjs/header'); ?>
2
3 <div class="card">
4 <div class="header-actions">
5 <h2>All Records</h2>
6 <button type="button" class="btn btn-success" onclick="openCreateModal()"> Create New Record
7 </div>
8
9
10 <table>
11 <thead>
12 <tr>
13 <th style="width: 50px;">ID</th>
14 <th style="width: 100px;">Image</th>
15 <th>Title</th>
16 <th>Author</th>
17 <th style="width: 250px;">Article Preview</th>
18 <th style="width: 200px;">Actions</th>
19 </thead>
20 <tbody>
21 <tr>
22 <td colspan="6" style="text-align: center; padding: 40px;">Loading...</td>
23 </tr>
24 </tbody>
25 </table>
26 </div>
27
28 <?php $this->load->view('crudjs/footer'); ?>

```

Gambar C.1.10 application/modules/Crudjs/views/footer.php, header.php, index.php

D. HASIL DAN DOKUMENTASI PROGRAM

1) Hasil Studi Kasus 1

Extra options

					id	title	author	article	image	created_at	updated_at
☐					4	Report	Zalfaa Ghaitsaa	This is a report	posts/1777713478_Screenshot__1_.png	2026-05-02 11:17:58	2026-05-02 11:17:58
☐					7	Report	Zalfaa Ghaitsaa	This is a report	posts/post_1777715546_69f5c95a4e27b.png	2026-05-02 11:52:26	2026-05-02 11:52:26
☐					8	Report 2	Zalfaa Ghaitsaa	This is a report	posts/post_1777715577_69f5c979a15e1.png	2026-05-02 11:52:57	2026-05-02 11:52:57
☐					9	Report 3	Zalfaa Ghaitsaa	This is a report	posts/post_1777715660_69f5c9cc7478e.png	2026-05-02 11:54:20	2026-05-02 11:54:20

☐ Check all With selected: Edit Copy Delete Export

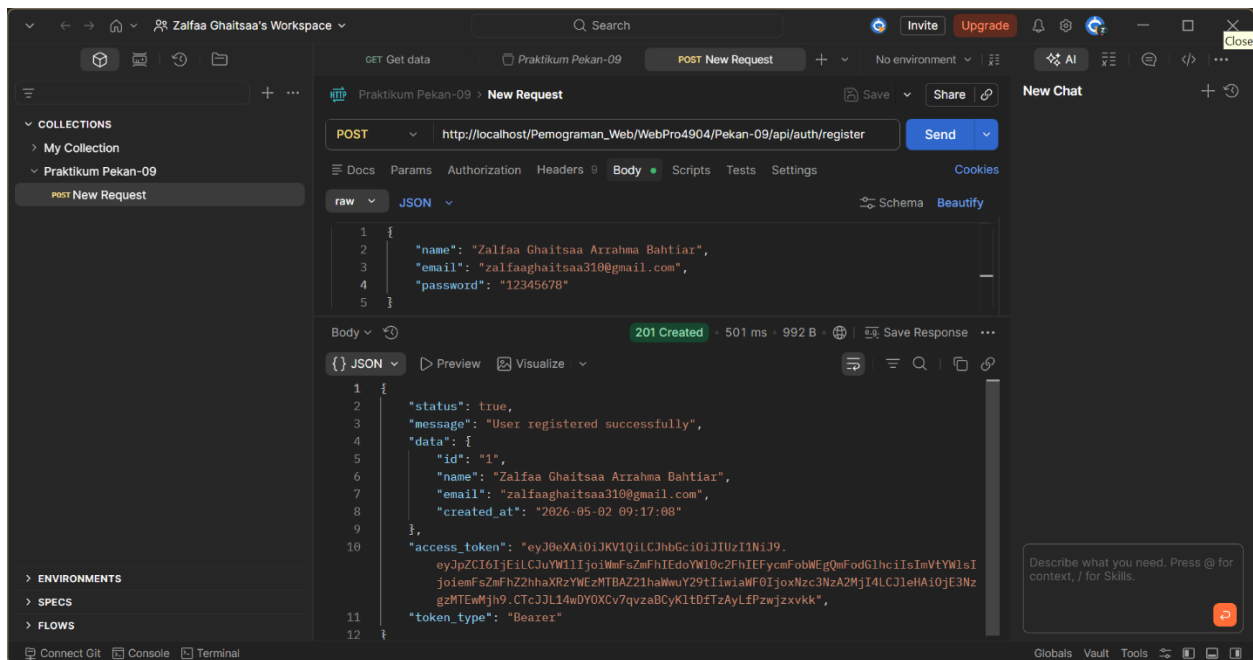
☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

					id	name	email	password	created_at	updated_at
☐					1	Zalfaa Ghaitsaa Arrahma Bahtiar	zalfaaghaitsaa310@gmail.com	\$2y\$10\$8T1Y.PmwTQ/1Kqu8h8TjO.PIDHvcVG15KNxrHrX6DQ...	2026-05-02 09:17:08	2026-05-02 09:17:08

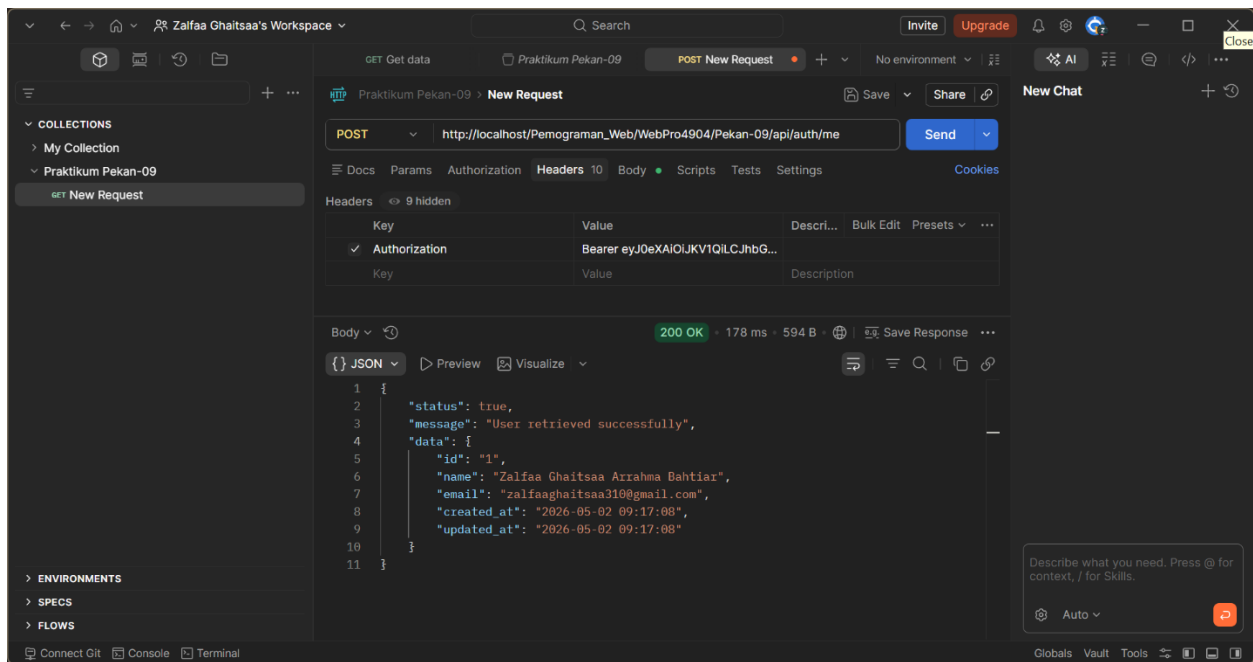
☐ Check all With selected: Edit Copy Delete Export

Gambar D.1.1 Tabel database

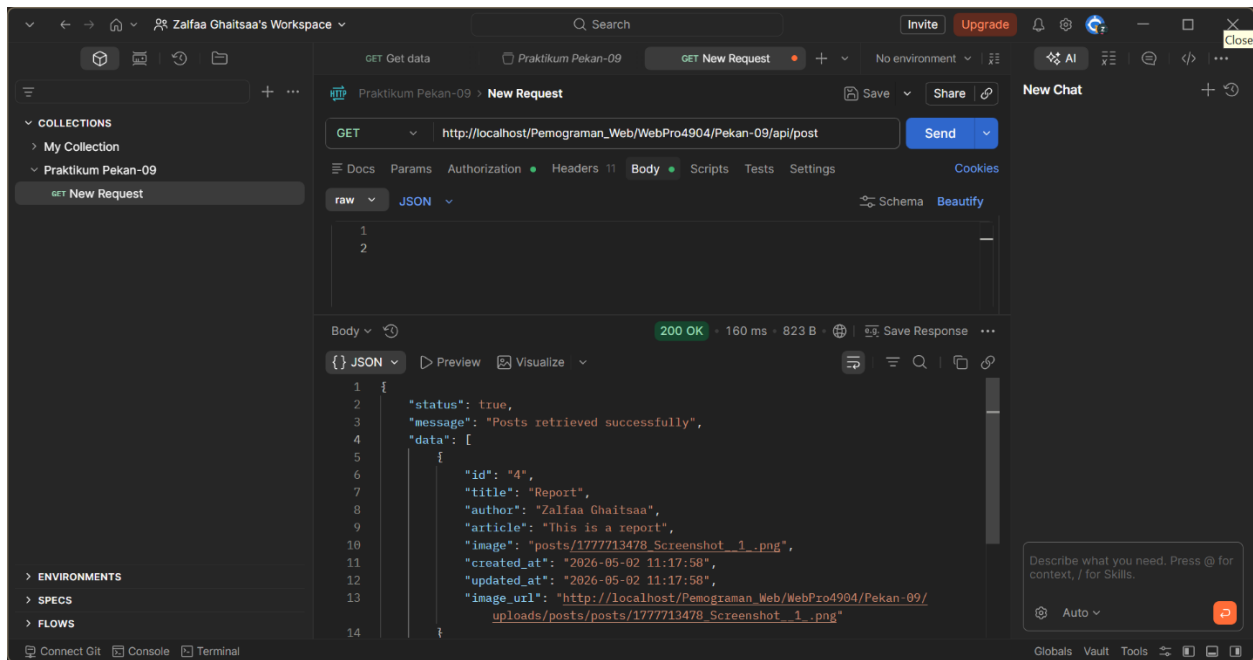


Gambar D.1.2 register di post man

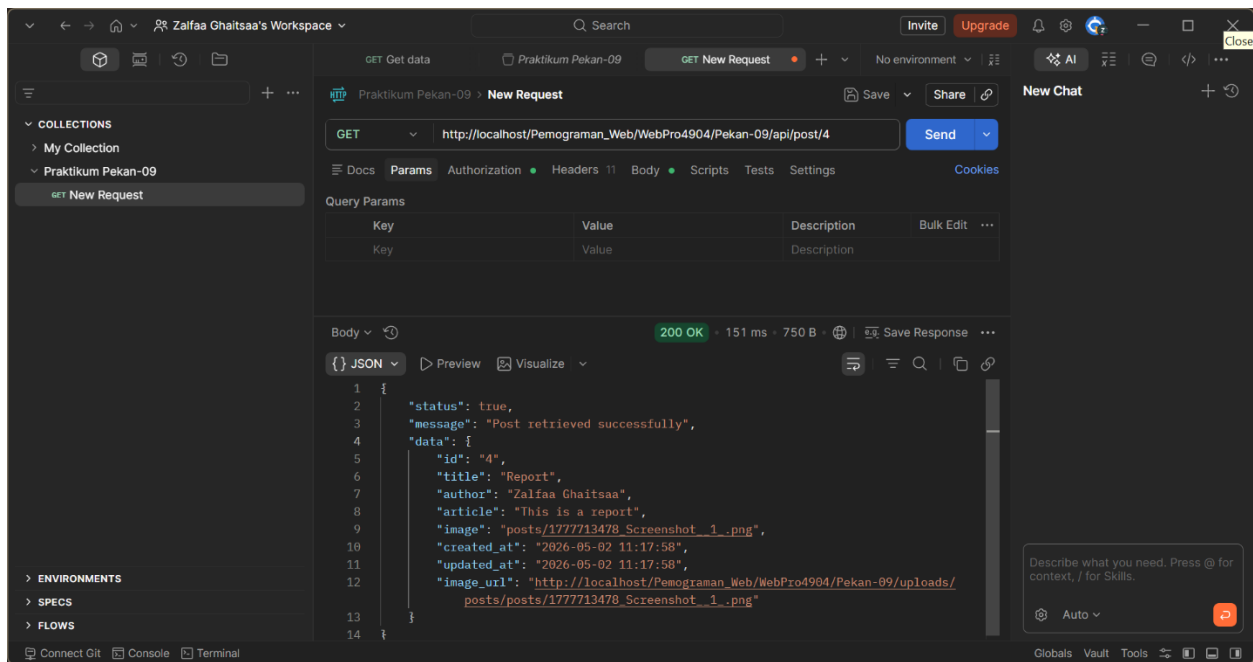




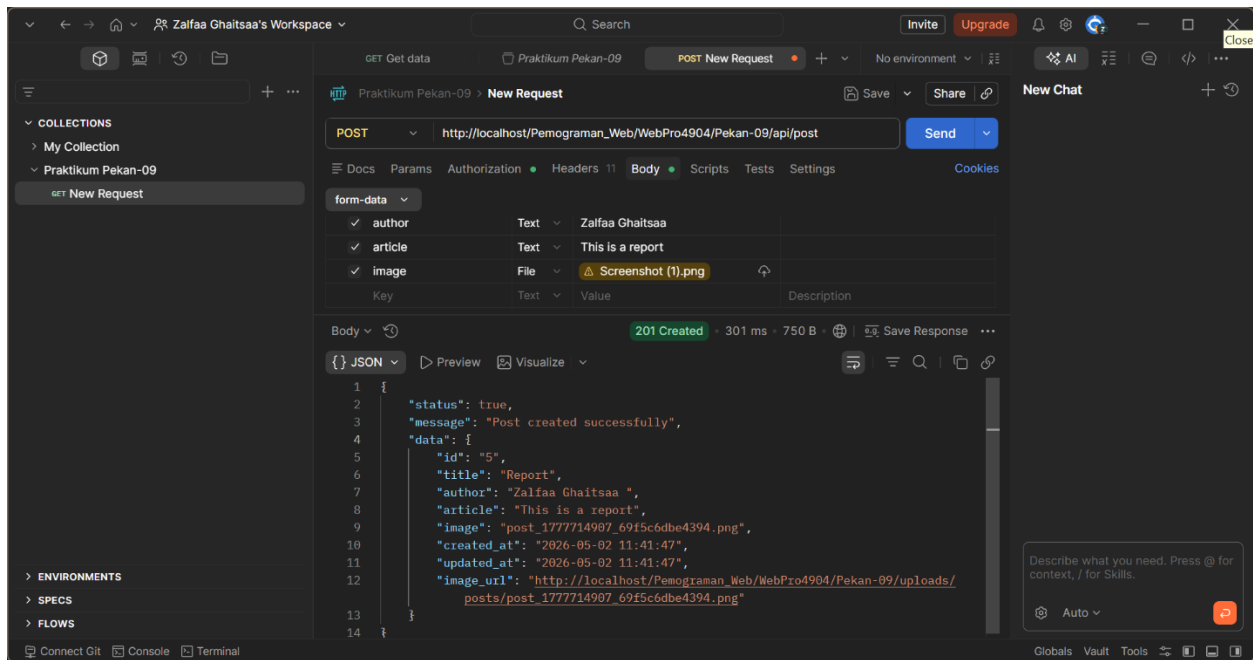
Gambar D.1.5 detail akun di post man



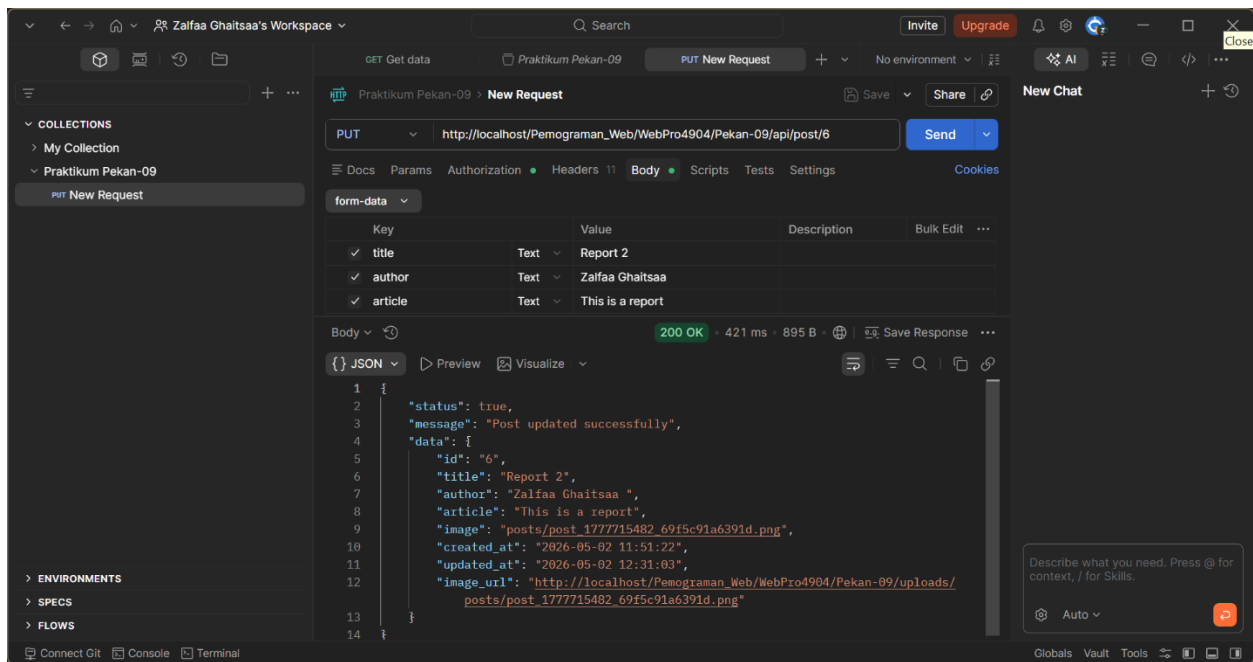
Gambar D.1.6 get all post di post man



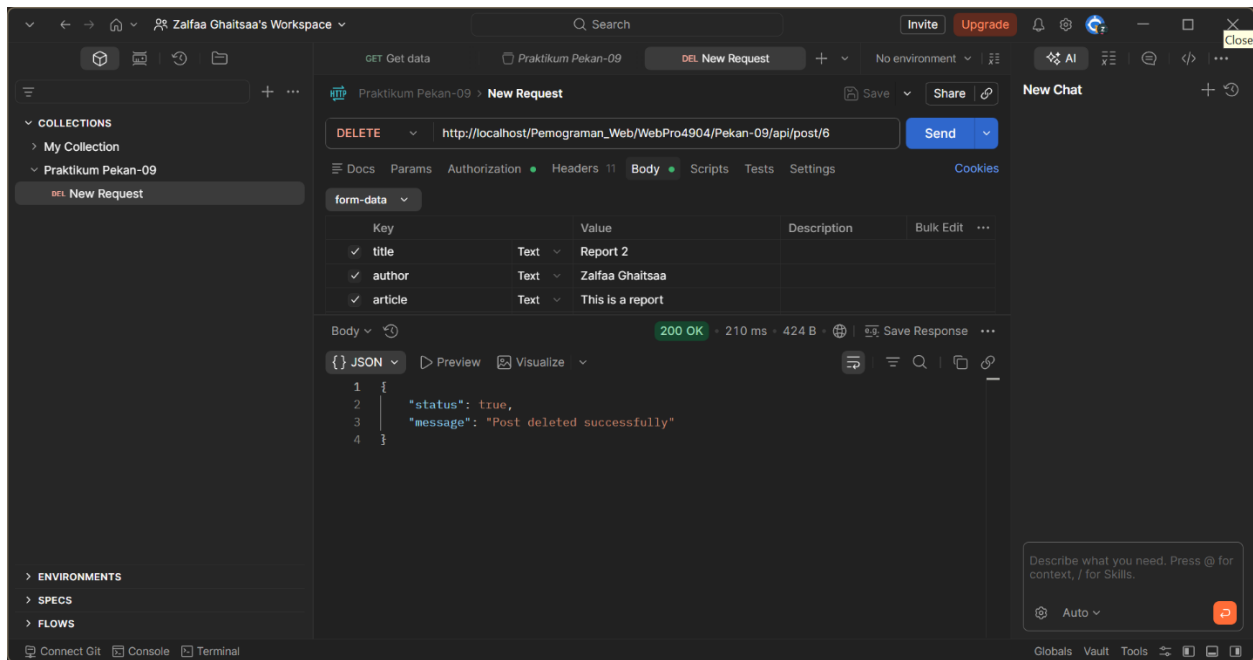
Gambar D.1.7 get single post di post man



Gambar D.1.8 create di post man



Gambar D.1.9 update di post man



Gambar D.1.10 delete di post man

E. ANALISIS DAN PEMABAHASAN

Praktikum ini difokuskan pada implementasi arsitektur Hierarchical Model-View-Controller (HMVC) yang bertujuan untuk meningkatkan efisiensi pengembangan sistem melalui modularitas. Berbeda dengan struktur MVC standar yang menyatukan seluruh komponen, HMVC memungkinkan setiap fitur seperti "Api", "Posts", atau "Crudjs" memiliki struktur MVC sendiri yang independen dan mandiri. Melalui pendekatan ini, pengelolaan kode menjadi lebih terorganisir karena setiap modul dapat dikembangkan, diuji, dan dipelihara tanpa harus mempengaruhi fungsionalitas modul lainnya secara langsung.

Pada aspek fungsionalitas, integrasi antara REST API dan AJAX memberikan dampak signifikan pada *User Experience* (UX) aplikasi. Penggunaan REST API memungkinkan pemisahan yang jelas antara backend yang mengolah data dengan frontend sebagai antarmuka pengguna. Saat melakukan operasi CRUD (Create, Read, Update, Delete), teknologi AJAX bekerja secara *asynchronous* untuk mengirimkan request di latar belakang tanpa memerlukan penyegaran halaman secara penuh (page refresh). Mekanisme ini memastikan aplikasi tetap responsif dan interaktif karena hanya data spesifik dalam format JSON yang ditransmisikan antara server dan browser.

Selain itu, aspek keamanan menjadi perhatian utama melalui implementasi JSON Web Token (JWT) untuk proteksi endpoint API. Setiap akses ke sumber daya yang sensitif, seperti pembuatan atau pembaruan postingan, memerlukan validasi token yang didapatkan melalui proses login. Pengujian yang dilakukan menggunakan Postman membuktikan bahwa sistem mampu merespons berbagai metode HTTP (GET, POST, PUT, DELETE) dengan status kode yang sesuai. Hal ini menjamin bahwa aliran data antar platform dapat berjalan dengan terstruktur, aman, dan memiliki skalabilitas yang baik untuk pengembangan sistem di masa depan.

F. KESIMPULAN

Secara keseluruhan, praktikum ini berhasil mengintegrasikan konsep modularitas HMVC dengan layanan RESTful API untuk menciptakan aplikasi web yang modern dan scalable. Mahasiswa mampu memahami bahwa penggunaan AJAX sangat efektif dalam menciptakan interaksi yang dinamis tanpa membebani server dengan reload halaman yang tidak perlu. Selain itu, penerapan JWT terbukti menjadi metode yang andal untuk menjaga keamanan data pada arsitektur sistem yang terpisah, sehingga aplikasi tidak hanya unggul dari sisi performa tetapi juga terjamin dari sisi autentikasi akses.

G. LAMPIRAN

<https://github.com/AriqAhmadNayaka/WebPro4904.git>